

Guangzhou Invitational Competition 2026

Skill 08: Mobile Applications Development

Functionality Module from Germany

1. Introduction

TaskWeave is an offline task management app designed to help users organize, schedule, and track tasks efficiently.

2. Description of Project and Tasks

You do not need to consider the program interface's aesthetics and the elements' position. The work can be inconsistent with the given wireframes, but you need to consider the ease of use.

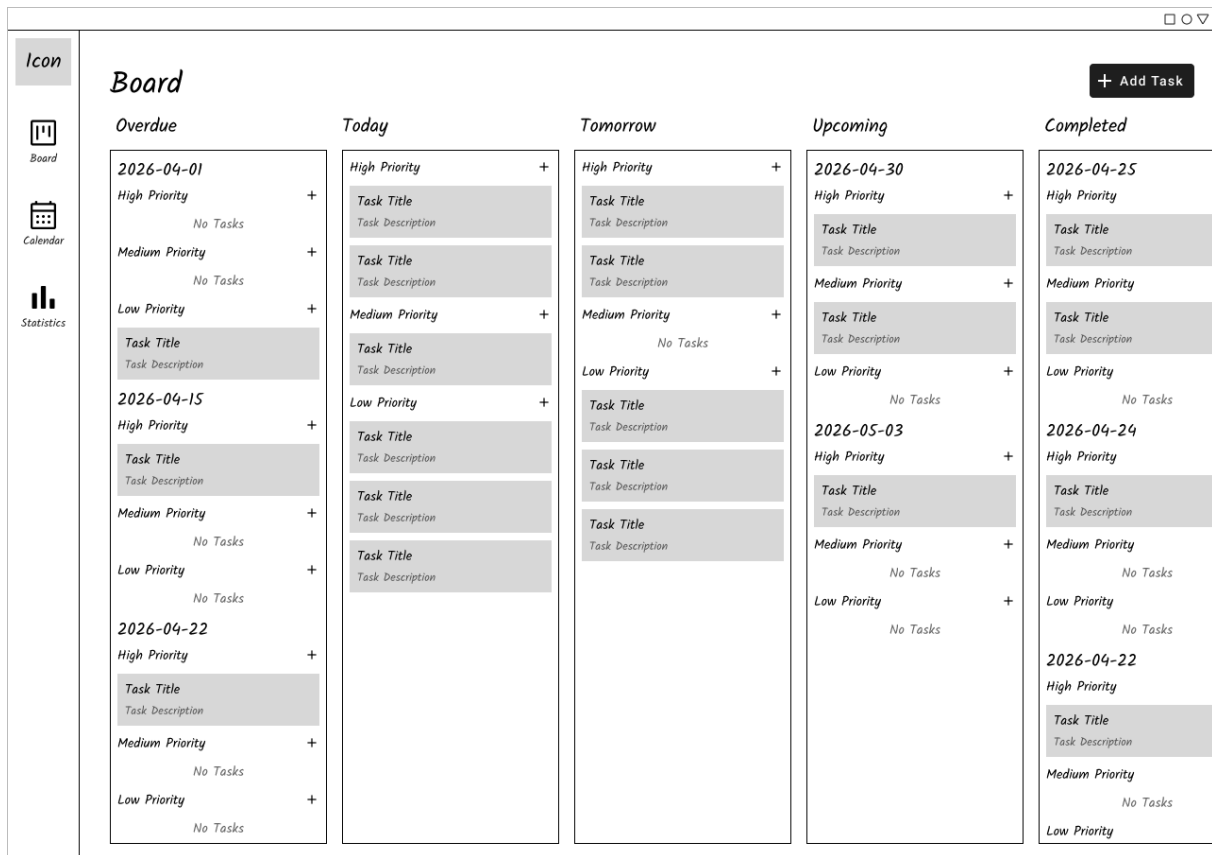
The app should be developed for the provided tablet emulator.

3. General Demands

1. Finish the function of this app according to the demands.
2. The app name must be "TaskWeave X" in the operating system, where XX is your workstation number.
3. The app must show the provided app icon `media_files/icons/taskweave_icon.png` in the operating system.
 1. When the app starts, the app icon is displayed to replace the default white splash screen.
4. The app starts on the Board Page.
5. All pages should contain a navigation bar on the left.
6. Left navigation bar:
 1. The app icon must be shown at the top.
 2. "Board" button with icon, can be clicked to go to the Board Page.
 3. "Calendar" button with icon, can be clicked to go to the Calendar Page.
 4. "Statistics" button with icon, can be clicked to go to the Statistics Page.
 5. Highlight the currently selected page.
7. Show the provided tasks when the app is first launched. The provided tasks must be editable and deletable, and new tasks can be created.
8. Create an app shortcut "Add Task" (which appears when long-clicking the app icon) that launches the app and opens the Task Dialog to create a new task.
9. Changes are persisted across app restarts.
10. After creating, editing, toggling completion, or deleting a task, a temporary message must appear with an "Undo" option that restores its previous state.

4. Page Demands

4.1. Board Page



4.1.1. Elements

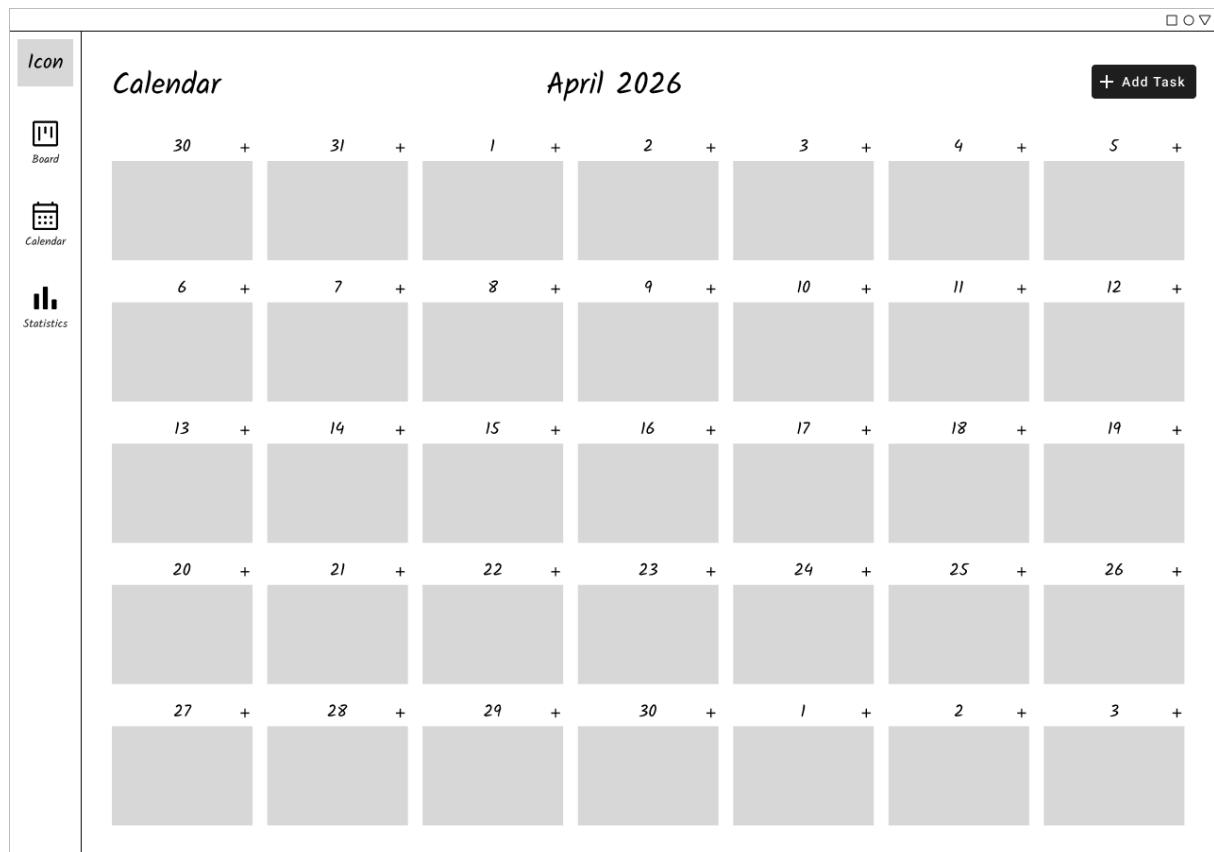
1. "Board" text
2. Add Task button
3. Horizontal list of task columns (scrollable if necessary):
 1. "Overdue"
 2. "Today"
 3. "Tomorrow"
 4. "Upcoming"
 5. "Completed"
4. Each column contains:
 1. Title
 2. Scrollable task list
5. Each priority group contains:
 1. Title ("High Priority", "Medium Priority", or "Low Priority")
 2. Color representing the priority
 3. Add Task button
 4. Tasks
6. Each task contains:
 1. Title (max 1 line)
 2. Optional short description (max 2 lines)

3. Repeating icon (if applicable)

4.1.2. Functional Requirements

1. Each task appears in a column based on its due date:
 1. Overdue: due date before today, not completed
 2. Today: due date is today, not completed
 3. Tomorrow: due date is tomorrow, not completed
 4. Upcoming: due date after tomorrow, not completed
 5. Completed: completed tasks
2. In the Today and Tomorrow columns:
 1. Tasks must be grouped by priority (High, Medium, Low).
 2. All groups are shown, even if they are empty.
 3. Tasks can be dragged and dropped to reorder them within the same column, even across priority groups (automatically changing their priority).
3. In the Overdue, Upcoming, and Completed columns:
 1. Tasks are grouped by due date (Overdue: oldest first, Upcoming: soonest first, Completed: most recent first), then by priority.
 2. Only due dates with tasks must be shown. However, for each shown date, all priority groups must be shown, even if they are empty.
 3. Tasks can be dragged and dropped to reorder them within the same column, even across due dates and priority groups (automatically changing their due date and priority).
4. Swiping from left to right on a task in the Today, Tomorrow, or Upcoming column moves its due date one day later.

4.2. Calendar Page



4.2.1. Elements

1. "Calendar" text
2. Year and month label
3. Add Task button
4. Monthly calendar grid
 1. Column header for each day of the week (Monday to Sunday)
 2. Row for every week of the month (number of rows can vary based on the month)
 3. Day Cell for each day of the month, and possibly some days from the previous and next month to fill the grid
5. Each Day Cell contains:
 1. Day number
 2. Add Task button
 3. Scrollable task list
6. Each task contains:
 1. Title (max 2 lines)
 2. Color representing the priority
 3. Repeating icon (if applicable)

4.2.2. Functional Requirements

1. Each Day Cell first contains open (non-completed) tasks in the same order as in the Today column of the Board Page (i.e., sorted by priority, then by custom reordering).

Completed tasks show up below the open tasks, also sorted by priority and custom reordering.

2. Completed tasks must be clearly distinguishable from open tasks.
3. By default, the current month is displayed and the current day is visually highlighted.
4. The user can swipe left or right to navigate between months. The year and month label must update accordingly.

4.3. Functional Requirements for both Board and Calendar Pages

1. For repeating tasks, all completed/overdue occurrences and all occurrences in the current and next month must be shown.
 1. Daily repeated tasks have one occurrence per day.
 2. Weekly repeated tasks have one occurrence per week on the same weekday as the original task.
 3. Monthly repeated tasks have one occurrence per month on the same day of the month as the original task.
 1. If a month does not have this day, use the last day of the month instead. For example, a task created on January 31st would have the following due dates for its monthly occurrences: January 31st, February 28th (or 29th in a leap year), March 31st, April 30th, ...
2. Clicking on a task toggles it between completed and not completed.
3. Long-pressing a task opens the Task Dialog to edit the task.
4. When a task gets created, edited, or toggled between open and completed, it must automatically move to the correct position. An animation is not necessary.
5. Clicking on an Add Task button opens the Task Dialog to create a new task.
 1. For Add Task buttons in a Day cell or priority group, the Task Dialog must open with the corresponding due date and/or priority preselected.

4.4. Task Dialog

Edit Task

Title

Text

Description

Text

Priority

High

Medium

Low

Repeats

Never

Daily

Weekly

Monthly

Due Date

S	M	T	W	T	F	S
0	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	0	0	0	0

4.4.1. Elements

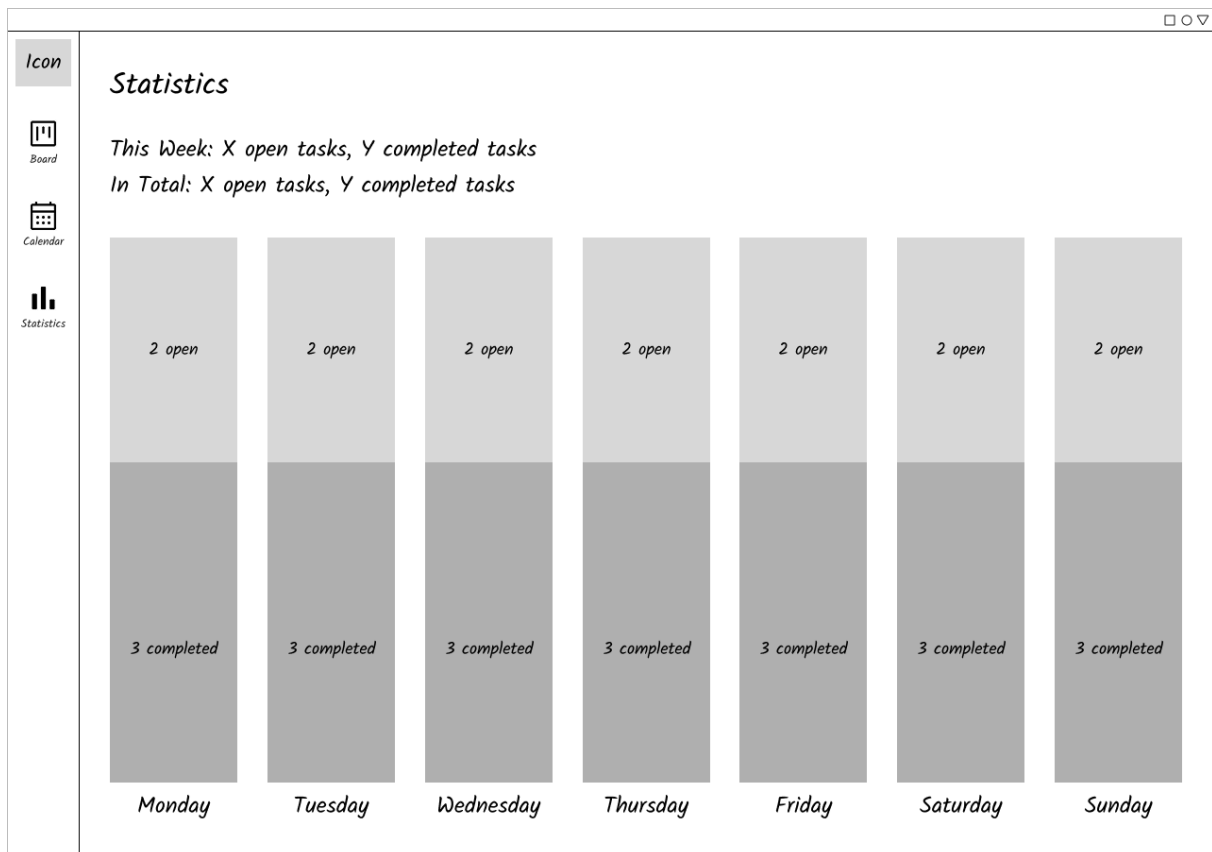
1. "Add Task" or "Edit Task" text depending on the context
2. Title text field with the text "Title"
3. Description text field with the text "Description"

4. Priority selector (High/Medium/Low) with the text "Priority"
5. Repeating selector (Never/Daily/Weekly/Monthly) with the text "Repeats"
6. Due Date picker with the text "Due Date"
7. Cancel Button
8. Create Button (only when creating a new task)
9. Save Button (only when editing an existing task)
10. Delete Button (only when editing an existing task)

4.4.2. Functional Requirements

1. The single-line Title text field validates that the title is not empty in real-time. If the title is empty, display a red error message "Title must not be empty" and disable the Create/Save Button.
2. The multi-line Description text field is optional and can be left empty.
3. The Due Date picker must support picking an arbitrary date from a calendar view (no text field, no additional dialog).
4. Unless otherwise noted, the Priority defaults to "Low", Repeating defaults to "Never", and Due Date defaults to one week from now.
5. The Priority, Repeating, and Due Date must always have a valid value selected.
6. Clicking the Cancel Button closes the dialog without saving changes.
7. Clicking the Create Button creates a new task with the specified properties and closes the dialog.
8. Clicking the Save Button applies changes to the existing task and closes the dialog.
9. Clicking the Delete Button deletes the existing task and closes the dialog.
10. When a repeating task is edited:
 1. The Repeating selector and Due Date picker must be disabled.
 2. When clicking the Save button, this and all future occurrences must be updated. Past occurrences must not be affected.
 3. When clicking the Delete button, the user must be prompted to choose whether to delete only the current occurrence or all future occurrences as well. Past occurrences must not be affected.

4.5. Statistics Page



4.5.1. Elements

1. "Statistics" text
2. Text "This Week: X open tasks, Y completed tasks", where X and Y are calculated based on the current week (Monday to Sunday)
3. Text "In Total: X open tasks, Y completed tasks", where X and Y are calculated based on all tasks (For repeating tasks, each occurrence is counted separately.)
4. Bar chart visualization for the tasks of the current week:
 1. one column for each day of the week (Monday to Sunday)
 2. each column contains:
 1. if there is at least one open task for this day:
 1. segment representing the number of open tasks
 2. label with the number of open tasks
 2. if there is at least one completed task for this day:
 1. segment representing the number of completed tasks
 2. label with the number of completed tasks
 3. weekday label

4.5.2. Functional Requirements

1. All numbers are calculated and updated when tasks are created, edited, or deleted.

5. Instructions to the Competitor

Your submission should contain the following top-level files and directories:

- the compiled app (APK or IPA file)
- a directory with the full source code
- optional: a README file containing notes for the marking

These files and directories should be pushed to the provided Git repository. Only the latest contents of the `main` branch will be taken into account.